a **6502 finite census: all completed arms**

L/BRK/budget	Periods k	Words tested	Exact self-writers
16/fait/128	1,2,3,4	4,311,810,304	0
16/fait/512	1,2,3,4	4,311,810,304	0
16/halt/128	1,2,3,4	4,311,810,304	0
16/halt/512	1,2,3,4	4,311,810,304	0
20/fait/128	1,2,3	16,843,008	0
20/fait/512	1,2,3	16,843,008	0
20/halt/128	1,2,3	16,843,008	0
20/halt/512	1,2,3	16,843,008	0
32/fait/128	1,2,3	16,843,008	0
32/fait/512	1,2,3	16,843,008	0
32/halt/128	1,2,3	16,843,008	0
32/halt/512	1,2,3	16,843,008	0

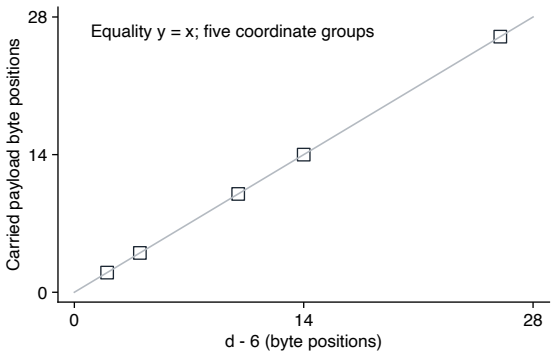
L = tape bytes; budget = execution steps; k = tiled-word period
fait = faithful BRK; halt = halt before BRK; exact output into zero B
Words tested sums k; each L16 k4 arm exhausts 256 to the fourth power words

b **8086 finite census: declared reset**

L/budget/k	Words tested	Exact output zero B	≥75% match in ≥8/16
16/128/1	256	0	0
16/128/2	65,536	20	29
16/512/1	256	0	0
16/512/2	65,536	29	29
32/128/1	256	0	0
32/128/2	65,536	26	36
32/512/1	256	0	0
32/512/2	65,536	26	25

L bytes; budget execution steps; k = word period; SP at ring end
Counts of nonzero words: exact output into all-zero byte partner B or ≥75% parent-byte match in ≥8/16 uniform-partner encounters

c **Constructed 6502: sampled carriage**



296 constructed copier/contract configurations; points coincide
d – 6 / configurations: 2/96; 4/40; 10/96; 14/32; 26/32
d = copy displacement in bytes
Carried: same-position parent byte in all 3 bodies×64 partner tests

d **S1: complete four-arm route ledger**

Arm L/budget	Worlds	Conf.	Open	Undet.	Loop (overlaps)
A 16/128	20	20	0	0	14
B 32/128	20	4	0	16	4
C 16/512	20	19	1	0	15
D 16/128	20	18	0	2	17

S1: 6502; Conf. = no partner fetch in 16/16 traces; Open = fetch ≥8/16
Undet. = no heritable copier detected; Loop = repeated instruction address
Loop overlaps routes; L bytes/budget steps; A–C faithful, D pre-BRK

e **Registered C2: complete sampling roles**

Sampling role	Sampled	Eligible	Elig. worlds
Final random cell	640	23	4
Final top-3 by abundance	240	10	4
First detected tape	62	9	9

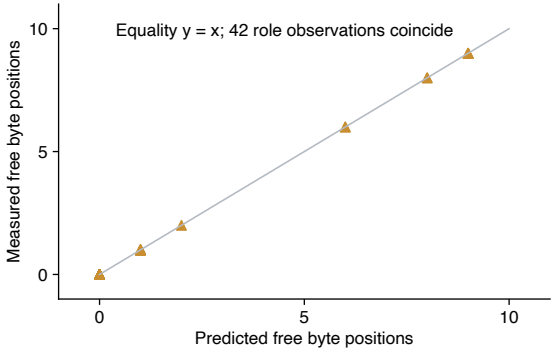
Eligible: 42 roles; 41 unique tapes; 11 worlds

6502 retention experiment C2: 80 S1 worlds; fixed-core certificate

Pointer randomization: unmeasured; not an 8086 S2 arm

Eligible = certified fixed-core, pure forward-copy family
942 roles; world counts overlap across roles (do not sum)

f **C2 exact capacities: all eligible roles**



Free byte positions: arbitrary alternatives jointly inherited with causal roots
Each exact-byte/core/ancestry check: 0/91,428 direct-case mismatches and 0/731,424 serial-record mismatches; 42 roles /41 tapes /11 worlds